

Primary results of NRG Oncology / NSABP B-43: Phase III trial comparing concurrent trastuzumab (T) and radiation therapy (RT) with RT alone for women with HER2-positive ductal carcinoma in situ (DCIS) after lumpectomy.



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Abstract

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Background: Preclinical studies report that T can boost RT effectiveness. The primary aim of this trial assessed the efficacy of concurrent T + RT vs RT alone in preventing recurrence of ipsilateral breast cancer, ipsilateral skin cancer, or ipsilateral DCIS (IBTR) in women with DCIS. **Methods:** Eligibility: Women ≥ 18 yrs, ECOG performance status 0 or 1, DCIS resected by lumpectomy, and clear margins. Whole-breast RT after randomization was with 25+ fractions or accelerated with 16-17 fractions. RT boost was allowed. Centralized HER2 testing and ER and/or PR were required before entry. Stratification was by menopausal status, adjuvant endocrine therapy plan, and nuclear grade. T was given at 8 mg/kg IV within 1 wk before and 5 days after RT began (Dose 1) and at 6 mg/kg IV 3 wks after Dose 1 (Dose 2). Definitive intent-to-treat primary analysis was to be conducted when either 163 IBTR events were recorded or when all accrued pts were on study for ≥ 5 yrs. **Results:** 2014 pts were randomized (11/9/08 to 12/8/14); 1998 (99.2%) had follow-up information. Median follow-up time on 12/31/19 was 79.2 mos. 2001 pts had RT information, 1965 (98.2%) completed RT: 988 (98.3%) in the RT arm and 977 (98.1%) in the RT+T arm. 996 pts had T compliance information in the RT+T arm, 939 (94.3%) completed two doses of T, 25 (2.5%) had one dose of T, and 32 (3.2%) did not receive T. At primary definitive analysis, 114 IBTR events occurred: 63 in the RT arm and 51 in the RT+T arm (HR = 0.81 [95% CI: 0.56-1.17], p-value = 0.26). 38 were invasive: 18 in the RT arm and 20 in the RT+T arm (HR = 1.11 [95% CI: 0.59-2.10], p-value = 0.74). 76 were DCIS: 45 in the RT arm and 31 in the RT+T arm (HR = 0.68 [95% CI: 0.43-1.08], p-value = 0.10). Annual IBTR event rates were 0.99%/yr in the RT group and 0.80%/yr in the RT+T group. There were 288 events of any kind [iDFS-DCIS] (DFS): 155 in the RT arm and 133 in the RT+T arm (HR = 0.84 [95% CI: 0.66-1.05], p-value = 0.13) and 48 deaths: 26 in the RT arm and 22 in the RT+T arm (OS HR = 0.85 [95% CI: 0.48-1.51], p = 0.59). The study did not reach the 163 protocol-specified events, so the definitive analysis was triggered by all pts having been on study for ≥ 5 years. **Conclusions:** The addition of T to RT did not achieve the protocol objective of 36% reduction in the IBTR rate but did achieve a modest, statistically non-significant reduction of 19%. Support: U10-180868, -180822, UG1-189867; Genentech. The

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